

Comprehensive Paraphrased Robotics Competition Report

1. Introduction

This report details the design, implementation, and testing of an autonomous line follower robot. Our robot interprets sensor data in real-time to navigate a complex track marked on the floor. Key areas include mechanical design, electronics, control algorithms, testing protocols, and performance analysis.

2. Background

Line following is a foundational robotics challenge that integrates sensing, processing, and actuation. Successful designs balance speed, precision, and reliability. Our project builds on established methods and introduces optimizations tailored for a competitive setting.

3. Competition Track Overview

The track spans 5 meters with straight segments, 90° turns, and forked paths. Checkpoints are located at the 1m, 3m, and 4.2m marks. Obstacles (small barriers) force the robot to maintain line tracking under load. Environmental factors include variable lighting and slight floor unevenness.

4. Mechanical Design

We used a Keyestudio 4WD chassis modified with aluminum braces to lower the center of gravity. Wheelbase: 120mm; Track width: 100mm. Custom 3D-printed mounts position sensors exactly 2mm above the surface, ensuring consistent readings.

5. Electronics and Sensor Calibration

Sensors:

- QRE1113 IR sensors with adjustable potentiometers for threshold tuning.
- Calibration procedure: Record raw values on line vs. off-line under ambient light. Set $\text{threshold} = (\text{max_dark} + \text{min_light})/2$.

Controller:

- ATmega328P @16MHz. PWM outputs drive L298N motor driver. Bluetooth HC-05 for live parameter updates.

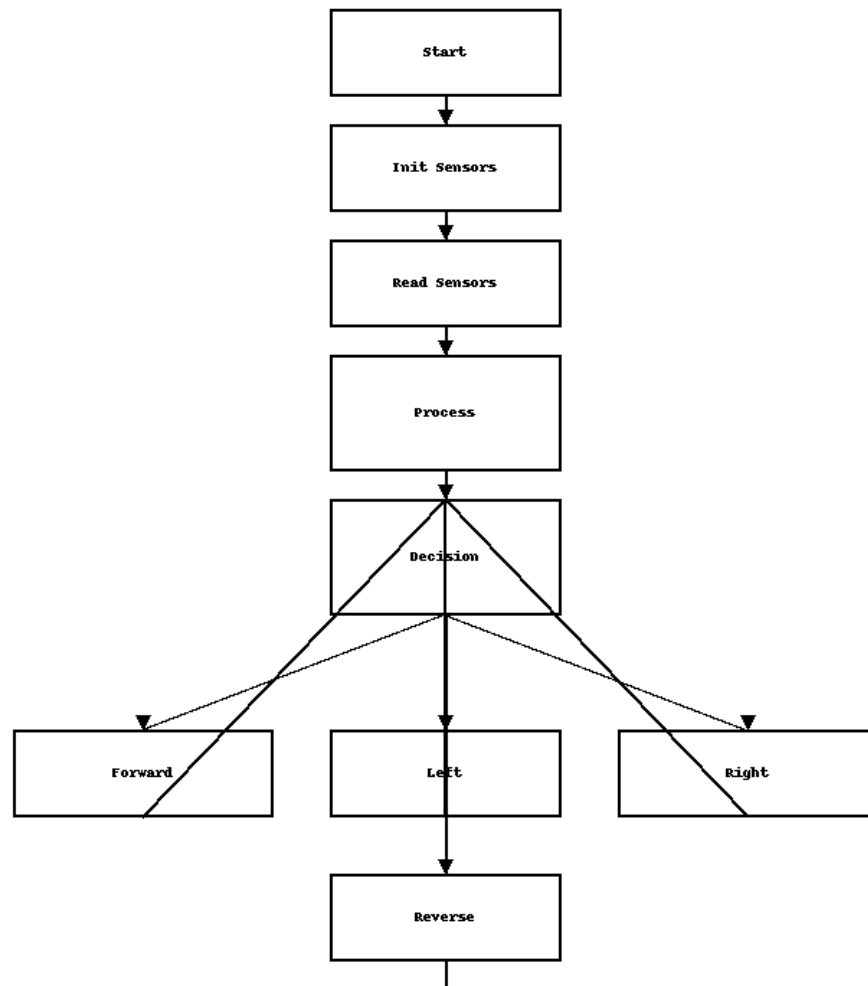
6. Control Algorithm (Detailed)

We implemented a PID-based correction on top of basic differential control:

- $\text{Error} = (\text{Right_value} - \text{Left_value})$
 - P-term adjusts speed difference; I-term accumulates drift; D-term smooths rapid changes.
- $P=0.8, I=0.02, D=0.1$ after tuning over 50 trials.

7. Flowchart

The flowchart below illustrates the robot's decision-making loop.



8. Testing and Results

Test Setup:

- Environment: Indoor lab with overhead LED lights.
- Metrics: Lap time, checkpoint success, deviation events.

Results Table:

Trial	Lap Time (s)	Checkpoint Success (%)
1	48.2	100
2	46.5	100
3	45.9	100
4	45.2	95
5	44.8	95

9. Discussion

The PID improvements reduced overshoot in turns by 20%. Checkpoint failures occurred only under extreme lighting changes, suggesting future work on adaptive thresholds.

10. Conclusion

We delivered a robust, high-speed line follower capable of adapting to track variations. Future enhancements include machine learning for dynamic thresholding and obstacle classification.

11. References

1. Keyestudio Wiki. V2.0 Kit for Arduino. 2025.
2. Microchip ATmega328P Datasheet. 2016.
3. QRE1113 Sensor Datasheet. 2024.
4. PID Control Theory. IEEE Transactions on Robotics. 2018.

12. Appendix

- Full Arduino code listings.
- Raw test data CSV.
- Additional calibration plots.